UNIT - IV

CAPITAL BUDGETING

Subject: Managerial Economics & Financial Analysis (23A52402a)

Syllabus at a glance

- Capital: nature, meaning, significance; types of working capital and components
- Sources of short-term and long-term capital
- Estimating working capital requirements
- Capital budgeting: features, proposals, methods and evaluation
- Pay Back, ARR, NPV, IRR methods (sample problems)

1. Introduction to Capital

1.1 Meaning

- **Capital** = the funds (money and assets) required to start, run and expand a business.
- Total capital is divided into **Fixed (long-term) capital** and **Working (short-term) capital**.

Fixed capital	Working capital
Invested in fixed assets: land, building, machinery	Invested in current assets: stock, debtors, cash
Long-term, not converted into cash quickly	Short-term, rotates in the operating cycle
Raised from shares, debentures, term loans	Raised from trade credit, bank overdraft, cash credit
Decision = capital budgeting	Decision = working capital management

1.2 Significance of capital management

- Ensures the **right amount of funds at the right time** at the lowest cost.
- Maintains **liquidity and solvency**, supports growth and increases the **value of the firm**.
- Avoids both shortage (insolvency risk) and excess (idle funds, low return).

2. Working Capital

2.1 Meaning

Gross working capital = total current assets. **Net working capital** = Current assets - Current liabilities.

- Current assets: cash, bank, marketable securities, debtors (receivables), bills receivable, inventory (raw material, WIP, finished goods), prepaid expenses.
- Current liabilities: creditors, bills payable, outstanding expenses, short-term loans, bank overdraft, tax payable.

2.2 Concepts

- **Gross concept** focuses on the optimum investment in current assets; **Net concept** shows liquidity position and the margin of safety for creditors.

2.3 Working capital cycle (operating cycle)



- The cycle repeats: cash buys raw material, then work-in-progress, finished goods, sales on credit (debtors), and back to cash.

Operating cycle = Inventory period + Receivable period - Payable period

2.4 Types of working capital

Basis	Types
By concept	Gross and Net working capital
By time	**Permanent (fixed / regular)**: minimum level always needed; **Temporary (variable)**: seasonal and special needs
Others	Reserve margin working capital; Initial working capital; Regular working capital

2.5 Importance and factors affecting working capital

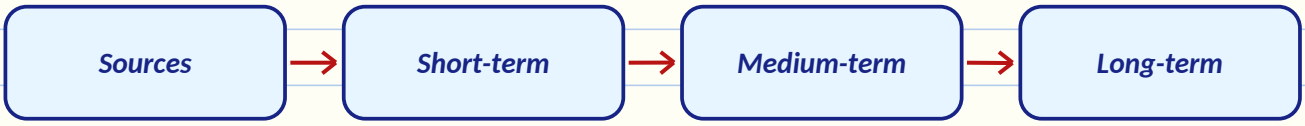
- Ensures smooth production, payment of wages / dues, goodwill with creditors, and ability to take advantage of discounts.
- Factors: **nature and size of business**, production cycle length, credit policy (sales and purchase), seasonal variations, growth and expansion, inflation, availability of raw

material, dividend policy, and level of competition.

2.6 Over-capitalisation vs Under-capitalisation of working capital

Excess working capital	Inadequate working capital
Idle funds, low return	Production interruptions
Wasteful spending, stock piling	Cannot pay creditors on time, loss of goodwill
Speculative activities	Cannot avail cash discounts, overtrading

3. Sources of Finance



3.1 Short-term sources (up to 1 year)

- **Trade credit** (suppliers allow time to pay); **Bank overdraft/ cash credit** (up to a limit); **Bills discounting**.
- **Commercial paper**, short-term loans, **advances from customers**, outstanding expenses, factoring of receivables, public deposits.

3.2 Medium-term (1 - 5 years)

- Term loans from banks, lease financing, **hire purchase**, public deposits, medium-term debentures.

3.3 Long-term sources (more than 5 years)

Source	Key points
Equity shares	Permanent capital, no fixed cost, owners' risk, voting rights
Preference shares	Fixed dividend, priority over equity, no voting
Debentures / bonds	Fixed interest, tax-deductible, no ownership
Term loans (banks, financial institutions)	Fixed repayment schedule; IDBI, SIDBI, NABARD
Retained earnings (ploughing back)	Internal source, no cost of issue
Venture capital, ECB, IPO / FPO	For start-ups, foreign funds, public issues
Leasing	Use of asset without owning; finance and operating lease

3.4 Debt vs Equity

Basis	Equity	Debt
Return	Dividend (not compulsory)	Interest (compulsory)
Risk to firm	Low	High (default risk)
Control	Dilutes control	No dilution
Tax	Dividend not deductible	Interest tax-deductible

4. Estimating Working Capital Requirements

4.1 Methods

- **Current assets holding period method**; **Operating cycle method**; **Percentage of sales method** (WC = 20-25% of sales); **Regression analysis**; **Forecast of components**.

4.2 Format

Particulars	Amount
Raw material stock (period x monthly cost)	A
Work-in-progress	B
Finished goods stock	C
Debtors (at cost)	D
Cash and bank balance	E
Total current assets	A + B + C + D + E
Less: Creditors, outstanding expenses	(F)
Net working capital	Total CA - F
Add: Contingency / safety margin (10%)	G

4.3 Solved Problem

Annual output 60,000 units. Raw material Rs. 20, labour Rs. 10, overheads Rs. 10 per unit. Raw material stock 1 month; WIP 1/2 month (100% material, 50% conversion cost); finished goods 1 month; debtors 2 months (at cost); creditors 1 month (raw material); cash balance Rs. 50,000. Estimate working capital.

- Monthly: Raw material = 12,00,000 / 12 = **1,00,000**; Labour = 50,000; Overheads = 50,000; total cost = 2,00,000.

Item	Working	Rs.

Raw material stock	1 month x 1,00,000	1,00,000
Work-in-progress	Material 50,000 + (50% x 25,000) labour + (50% x 25,000) OH	75,000
Finished goods	1 month x 2,00,000	2,00,000
Debtors	2 months x 2,00,000	4,00,000
Cash balance	Given	50,000
Total current assets		**8,25,000**
Less: Creditors	1 month x 1,00,000	(1,00,000)

Net working capital = 8,25,000 - 1,00,000 = Rs. 7,25,000

- With 10% contingency: 7,25,000 + 72,500 = Rs. 7,97,500.

5. Capital Budgeting

5.1 Meaning

Capital budgeting is the process of **evaluating and selecting long-term investment projects** that will generate returns over several years. (Charles Horngren: "Long-term planning for making and financing proposed capital outlays.")

5.2 Nature / Features

- Involves **large funds** and **long-term commitment**; benefits come in future years.
- **Irreversible** (high cost of reversal); involves **risk and uncertainty**; affects the long-term **profitability and growth**.
- Decision is strategic and made by top management.

5.3 Significance

- Controls **large investments**, avoids over / under investment, and ensures **wealth maximisation**.
- Helps in choosing among alternative projects and in rationing scarce capital.

5.4 Types of capital budgeting proposals

- **Replacement proposals** (replace old machine); **Expansion proposals** (increase capacity); **Diversification** (new products / markets); **New project** (new plant); **Mandatory / welfare** (safety, pollution control).
- **Independent**, **mutually exclusive** (choose one only), and **contingent** proposals.

5.5 Process

Idea generation → Screening → Evaluation → Selection → Implement & review

5.6 Methods of evaluation

Evaluation → Traditional → Modern (DCF)

Traditional (non-discounting)	Modern (discounted cash flow)
Payback Period	Net Present Value (NPV)
Accounting Rate of Return (ARR)	Internal Rate of Return (IRR)
	Profitability Index (PI)

6. Payback Period Method

6.1 Meaning

The time required to **recover the initial investment** from the project's cash inflows.

Shorter payback = better project.

Payback (even inflows) = Initial investment / Annual cash inflow

- For uneven inflows: add cumulative inflows and interpolate: **Payback = Years before recovery + (Unrecovered amount / Cash inflow of that year)**.

6.2 Merits and Demerits

Merits	Demerits
Simple, easy to understand	Ignores cash flows after payback
Focus on liquidity and risk	Ignores time value of money
Useful in uncertain / fast changing industries	Ignores profitability, arbitrary cut-off

Problem 1 (even inflows)

Investment Rs. 1,00,000; annual cash inflow Rs. 40,000 for 4 years.

Payback = 1,00,000 / 40,000 = 2.5 years

Problem 2 (uneven inflows)

Investment Rs. 50,000; inflows: Yr1 15,000; Yr2 20,000; Yr3 25,000; Yr4 10,000.

Year	Inflow	Cumulative
1	15,000	15,000
2	20,000	35,000
3	25,000	60,000

- Recovered during year 3: unrecovered after 2 years = 15,000; payback = 2 + (15,000 / 25,000) = **2.6 years**.

7. Accounting Rate of Return (ARR)

7.1 Meaning

ARR (Average Rate of Return) measures the **average annual accounting profit as a percentage of investment**. Accept if ARR > required rate.

ARR = (Average annual profit after tax and depreciation / Investment) x 100

- Investment may be **initial investment** or **average investment** = (Initial cost + Scrap value) / 2.

7.2 Merits and Demerits

- Merits: simple, uses accounting data, considers profits of the whole life.
- Demerits: **ignores time value of money**, uses profit and not cash flow, and no clear cut-off rate.

Problem 3

Machine cost Rs. 1,00,000, scrap value Rs. 20,000, life 4 years. Cash profit before depreciation is Rs. 40,000 each year.

- Annual depreciation = (1,00,000 - 20,000) / 4 = 20,000; profit after depreciation = 40,000 - 20,000 = **20,000**.

ARR on initial investment = 20,000 / 1,00,000 = 20%

ARR on average investment = 20,000 / 60,000 = 33.3% [Average investment = (1,00,000 + 20,000)/2]

8. Net Present Value (NPV)

8.1 Meaning

NPV = **Present value of cash inflows - Present value of cash outflows (initial investment)**, discounted at the cost of capital.

NPV = [C1/(1+k) + C2/(1+k)^2 + ... + Cn/(1+k)^n] - Initial investment

- **Decision rule**: NPV > 0 accept; NPV < 0 reject; NPV = 0 indifferent. For many projects choose the highest NPV.
- Present value factor = 1 / (1 + k)^n.

8.2 Merits and Demerits

- Merits: **considers time value of money**, all cash flows, gives absolute value addition (wealth maximisation).
- Demerits: needs a **correct discount rate**, more complex; favours larger projects; not easy for non-experts.

Problem 4

Initial investment Rs. 1,00,000; cost of capital 10%; inflows: Yr1 40,000; Yr2 50,000; Yr3 40,000; Yr4 30,000. PV factors at 10%: 0.909, 0.826, 0.751, 0.683.

Year	Cash inflow	PV factor	PV (Rs.)
1	40,000	0.909	36,360
2	50,000	0.826	41,300
3	40,000	0.751	30,040
4	30,000	0.683	20,490
Total PV of inflows			1,28,190

NPV = 1,28,190 - 1,00,000 = + Rs. 28,190 (Accept the project)

Profitability Index (PI)

PI = PV of inflows / Initial investment = 1,28,190 / 1,00,000 = 1.28

- PI > 1: accept; PI < 1: reject. Useful for **capital rationing**.

9. Internal Rate of Return (IRR)

9.1 Meaning

IRR is the **discount rate at which NPV = 0** (PV of inflows = PV of outflows). It is the project's own expected rate of return.



NPV falls as discount rate rises.
IRR = rate at which NPV = 0
Accept if IRR > cost of capital.

- **Decision rule**: accept if IRR > cost of capital (required rate); reject if IRR < cost of capital.

9.2 Computation

- **Even cash inflows**: Factor = Investment / Annual inflow; look up in the annuity table for n years.
- **Uneven inflows**: trial and error with two rates; use interpolation.

IRR = Lower rate + [NPV at lower rate / (NPV at lower - NPV at higher)] x (Higher - Lower rate)

Problem 5 (even inflows)

Investment Rs. 1,00,000; annual inflow Rs. 40,000 for 4 years.

- Factor = 1,00,000 / 40,000 = 2.5. Annuity factor for 4 years: at 21% = 2.540; at 22% = 2.494.

IRR = 21 + (2.540 - 2.500) / (2.540 - 2.494) = 21 + 0.87 = *21.87%*

Problem 6 (uneven inflows - same cash flows as Problem 4)

Rate	PV of inflows	NPV
20%	1,05,671	+ 5,671
25%	96,768	- 3,232

IRR = 20 + [5,671 / (5,671 + 3,232)] x 5 = 20 + 3.18 = *23.18%*

- IRR (23.18%) is greater than the cost of capital (10%), so **accept**.

9.3 Merits and Demerits

- Merits: considers time value, gives a percentage return, no need to know the cost of

capital in advance.

- Demerits: **multiple IRRs** with non-conventional cash flows; assumes reinvestment at IRR; difficult; may conflict with NPV for mutually exclusive projects.

10. Comparison and Revision

Method	Time value?	Decision rule
Payback	No	Lower payback is better than the standard
ARR	No	Higher than minimum required rate
NPV	Yes	NPV > 0
IRR	Yes	IRR > cost of capital
PI	Yes	PI > 1

NPV vs IRR: NPV gives the **absolute rupee gain** (better for ranking mutually exclusive projects); IRR gives a **percentage return**. They can conflict when project sizes or timing of cash flows differ.

Other concepts

- **Cost of capital** : minimum return required by investors (WACC). **Capital rationing** : limited funds, so projects are ranked by PI.
- **Discounted payback** : payback using discounted cash flows. **Risk analysis** : sensitivity analysis, scenario, certainty equivalent, and risk-adjusted discount rate.
- **Cash flow after tax** = Profit after tax + Depreciation (non-cash expense).

11. Time Value of Money and Cost of Capital

11.1 Time value of money

- Rs. 1 today is worth more than Rs. 1 in future because of earning capacity, inflation and risk.

Future value FV = P x (1 + r)^n ; Present value PV = FV / (1 + r)^n

Example: Rs. 1,000 at 10% for 3 years: FV = 1,000 x (1.10)^3 = **Rs. 1,331** . PV of Rs. 1,331 due in 3 years at 10% = Rs. 1,000.

- **Annuity** : equal payments each year; PV of annuity uses the cumulative factor table.

11.2 Cost of capital

- Cost of capital = the minimum rate of return a firm must earn to satisfy its investors; it is the **cut-off rate** for accepting projects.

Source	Cost formula
Debt (after tax)	Kd = Interest rate x (1 - tax rate)
Preference capital	Kp = Preference dividend / Price
Equity (dividend growth)	Ke = D1 / P0 + g
Retained earnings	Same as the cost of equity

WACC = (Weight of equity x Ke) + (Weight of debt x Kd) + ...

Example: Debt Rs. 4,00,000 at 10% interest (tax 30%) so Kd = 7%; Equity Rs. 6,00,000 with D1 = 5, P0 = 50, g = 5% so Ke = 10% + 5% = 15%.

WACC = (6,00,000 x 15% + 4,00,000 x 7%) / 10,00,000 = (90,000 + 28,000) / 10,00,000 = 11.8%

12. More Solved Problems

Problem 7: Choosing between two projects

Projects X and Y each cost Rs. 60,000; cost of capital 10%. Inflows - X: 20,000; 25,000; 30,000; 20,000. Y: 30,000; 30,000; 20,000; 10,000. (PV factors 0.909, 0.826, 0.751, 0.683)

Item	Project X	Project Y
PV of inflows	18,180 + 20,650 + 22,530 + 13,660 = 75,020	27,270 + 24,780 + 15,020 + 6,830 = 73,900
NPV	+ 15,020	+ 13,900
PI	1.25	1.23
Payback	2 + 15,000/30,000 = 2.5 years	2 years

- Y recovers money faster (liquidity), but **X has higher NPV and PI**, so choose **Project X** (NPV is the best criterion for wealth maximisation).

Problem 8: Discounted payback (data of Problem 4)

- Cumulative PV: Yr1 36,360; Yr2 77,660; Yr3 1,07,700. Unrecovered after 2 years =

22,340.

Discounted payback = 2 + 22,340 / 30,040 = 2.74 years (against simple payback of 2.25 years)

Problem 9: Working capital by percentage of sales

Annual sales Rs. 24,00,000; working capital needed is 25% of sales.

Working capital = 24,00,000 x 25% = Rs. 6,00,000

Problem 10: Operating cycle method

Raw material storage 30 days; WIP 15 days; finished goods 20 days; debtors 45 days; creditors 30 days. Annual cost of production Rs. 36,50,000 (365 days).

Operating cycle = 30 + 15 + 20 + 45 - 30 = 80 days

Daily cost = 36,50,000 / 365 = Rs. 10,000, so Working capital = 80 x 10,000 = Rs. 8,00,000

Problem 11: Decision on a replacement proposal

Old machine: operating cost Rs. 50,000 per year; new machine costs Rs. 80,000 (5-year life, no scrap) and cuts operating cost to Rs. 30,000. Cost of capital 10% (annuity factor for 5 years = 3.791).

- Annual saving = 50,000 - 30,000 = 20,000. PV of savings = 20,000 x 3.791 = 75,820.

NPV = 75,820 - 80,000 = - Rs. 4,180 (reject the replacement)

Problem 12: NPV with scrap value

Investment Rs. 2,00,000; annual inflow Rs. 70,000 for 4 years; scrap value Rs. 20,000 at the end of year 4; cost of capital 12%. (Annuity factor 3.037; PV factor yr4 = 0.636)

- PV of annual inflows = 70,000 x 3.037 = 2,12,590; PV of scrap = 20,000 x 0.636 = 12,720.

NPV = 2,12,590 + 12,720 - 2,00,000 = + Rs. 25,310 (accept)

13. Quick Revision Sheet

Fixed capital = capital budgeting (long-term). Working capital = day-to-day (short-term). Net WC = CA - CL.
Payback = Investment / inflow. ARR = Avg profit / Investment. NPV = PV inflows - PV outflows. IRR = rate where NPV = 0. PI = PV inflows / investment.

- Remember the **investment decision rules**: NPV > 0, IRR > k, PI > 1, shorter payback, higher ARR.
- Short notes to practise: operating cycle, capital rationing, WACC, mutually exclusive projects, leasing, factoring.

14. Important Exam Questions

- Define working capital. Explain its types, components and factors affecting it. (10 M)
- Discuss the short-term and long-term sources of finance. (10 M)
- Calculate working capital from the data of cost and holding periods. (10 M)
- What is capital budgeting? Explain its features, significance and process. (10 M)
- Explain the payback, ARR, NPV and IRR methods with merits and demerits. (10 M)
- Problems: Payback, ARR, NPV, PI and IRR for two projects; choose the better.